

The Hydra Family Through the Toolkit

Hydra, Antihydra and Fenrir as one system — program step 1 of the collection’s research plan

1. Overview

This report applies the toolkit developed on machine 1 (see the meta report and the machine reports in this collection) to the closest catalogued relatives: the Busy Beaver cryptids **Hydra** (BB(2,5)) and **Antihydra** (BB(6)), which iterate the same value map with mirrored halting conditions, and the FRACTRAN holdout family **Fenrir**, which turns out to be the same machine in base 5. All three are *walk-absorption* cryptids in the taxonomy of the meta report (§6.1) — a different halting-condition type from machine 1, which is exactly why it was chosen: to see which tools survive the type change.

Results: the implementations are verified against every trajectory the bbchallenge wiki publishes, including two affine conjugacies (Section 2); the family provably cannot cycle (T1); a **q-adic branch-memory theorem** (T2) explains structurally why congruence invariants cannot decide these machines — the sharpest “what makes them hard” statement in this collection; the even-run acceleration is exact (T3); the golden-ratio absorption model is made rigorous about itself (T4); and the branch statistic matches Geom(1/2) to three decimals (P4). Deep runs reproduce and extend the wiki’s checkpoints. The open cores remain open, as expected; what changed is that the reasons are now theorems.

2. One family, three machines

The value map is $H(n) = \text{floor}(3n/2)$: $H(2m) = 3m$, $H(2m+1) = 3m + 1$. Each machine couples it to a counter that walks +2 on one parity and −1 on the other, halting when the walk would go below zero:

machine	domain	start	counter walk	halts when	status
Hydra	BB(2,5)	$n_0 = 3$	+2 on odd, −1 on even	$\#even > 2 \cdot \#odd$	undecided
Antihydra	BB(6)	$n_0 = 8$	+2 on even, −1 on odd	$\#odd > 2 \cdot \#even$	undecided
Fenrir	FRACTRAN-2 2	$y_0 = 1$	+2 on odd, −1 on even	$x = 0$ at even y	probv. non-halting

Fenrir’s value map is $y \rightarrow 5 \cdot \text{floor}(y/2) + 2 \cdot [y \text{ even}]$: the same divide-by-2, multiply-by-q skeleton with $q = 5$. Everything below is proved for $q = 3$ and, where stated, for $q = 5$ verbatim.

Fidelity. The implementations reproduce every published trajectory: Antihydra’s $A(0,4) \rightarrow \dots \rightarrow A(9,86)$ with the conjugacy $b = n - 4$; Hydra’s $C(3,0) \rightarrow \dots \rightarrow C(78,6)$ with the conjugacy $N = 3n - 6$ (2,000 steps verified against the TM-form map $(3N+6)/2$, $(3N+3)/2$); and Fenrir’s $S(0,1) \rightarrow \dots \rightarrow S(4,35)$. Both conjugacies appear to be unstated on the wiki; they pin the one-variable forms to the machine-level forms exactly.

3. Theorems

3.1 T1: the family cannot cycle

$H(n) > n$ for every $n \geq 2$, and Fenrir's map exceeds y for $y \geq 2$ (the small values escape explicitly: $1 \rightarrow 0 \rightarrow 2 \rightarrow 7$). So all three machines halt or escape to infinity; periodic behavior is impossible. For machine 1 this trichotomy collapse required the potential theorems; here it is immediate — the value itself is the potential. The Skelet #1 scenario is excluded across the whole family.

3.2 T2: q-adic branch memory — why congruence invariants are blind

Theorem. For the Hydra map, $n_t \bmod 3^k$ is an explicit function of the last k parities $(p_{t-k}, \dots, p_{t-1})$ and of nothing older. *Proof.* $n_t = 3 \cdot \text{floor}(n_{t-1}/2) + p_{t-1}$, and halving is invertible mod 3^k , so mod 3^k the recursion is $r \rightarrow 3(r - p)/2 + p$: the factor 3 pushes the unknown older state up one 3-adic digit per step, annihilating it after k steps. The same holds mod 5^k for Fenrir. (30,000 randomized checks each, plus a direct check that the start residue does not influence the result.)

Why this matters. It is the sharpest structural answer this collection has to “what makes these machines undecidable in practice.” The halting condition is a *cumulative count* of branches; the theorem says the value's residue to any fixed modulus built from q carries only a *bounded window* of branch history. So no congruence invariant of the value can track the walk: the q -adic side of n stores the recent past, the 2-adic side decides the future, and the halting condition lives in neither. On machine 1 the analogous fact was an empirical search result (no separating modulus $\leq 256 / \leq 628$); here, for the value alone, it is a theorem with a two-line proof. It also explains the community's experience that regular-language deciders fail on exactly these machines: a DFA over digits is a bounded-memory observer of a process whose halting depends on an unbounded count.

3.3 T3: exact acceleration

$H^S(2^S t) = 3^S t$, so a maximal run of even steps collapses to one jump with certificate s . An orbit is then a sequence of **blocks** — one odd step followed by v even steps, v the 2-adic valuation of the odd step's output — with per-block walk increment $2 - v$ for Hydra's counter and $2v - 1$ for Antihydra's. This is the family's version of the cascade batching of machine 1, verified block-runner against single steps.

3.4 T4: the absorption model, exact

For the model walk (+2 or -1, probability 1/2 each) the probability of ever reaching -1 from height h is q^{h+1} where q is the root in $(0,1)$ of $q = (1 + q^3)/2$:

$$q^3 - 2q + 1 = 0, \quad q = (\sqrt{5} - 1)/2 \approx 0.618034,$$

the golden-ratio conjugate — the constant behind every probvious verdict on this family. Verified as the minimal solution of the hitting system by monotone iteration (agreement to 10^{-6} , and q^{h+1} matches the iterated values for $h \leq 40$). The statement is exact *about the model*; whether the parity stream justifies the model for the one orbit that matters is the single-orbit gap, as everywhere in this collection.

4. Measurements

P4 (branch statistic). Over 200,000 blocks of the pure map, the valuation distribution matches $\text{Geom}(1/2)$ to three decimals ($v = 0: 0.5000; 1: 0.2498; 2: 0.1259; 3: 0.0621; \dots$), with $E[v] = 0.998$ against the model's 1. The family thus shares the collection's geometric branch law — the same law the Collatz heuristic assumes for the valuation of $3n + 1$. P4 now holds in every machine of this collection.

Deep runs. Hydra ran 2,000,000 pure-map steps (counter $b = 1,003,573$, exactly the drift $b \approx t/2$ the wiki reports at 4M, no halt); Antihydra 2,000,000 (counter $a = 996,805$, minimum $a = 2$ — it never approached 0, hence never halted); Fenrir 1,500,000 ($x = 751,413$, no halt). Values reached $10^{350,000} - 10^{600,000}$. All reproduce the wiki's reported drifts.

5. What transferred, what did not

tool	outcome	note
no-cycle proof (P6)	transferred trivially	the value is its own potential; T1
exact acceleration (P1/P5 machinery)	transferred	T3 blocks = the cascade batching, with certificates
branch statistic (P4)	transferred	$\text{Geom}(1/2)$ to three decimals; now seen in all machines
congruence analysis (P3)	transferred and upgraded	from a search result to a theorem: T2 proves no congruence invariant of the value can decide halting
explicit halting set (P2)	does not apply	walk-absorption type: the halting event is cumulative, not positional — as the taxonomy predicted
P8 accounting	applies via T4	per-opportunity probabilities q^{h+1} ; Σp convergent along the drifting walk

The taxonomy of the meta report made two predictions for this type: that P2's sparse-set machinery would not apply, and that the walk model would govern the risk accounting. Both held. The genuine surprise is T2: on the coincidence-type machines the impossibility of congruence separation was only ever an empirical search result, but on the absorption type it is a provable structural fact. Whether a T2-style theorem exists for machine 1 — some precise sense in which their orbit residues carry only bounded history — is now the sharpest question the transfer raises.

6. Status and verification

check	scope	result
implementations vs wiki trajectories	Antihydra $A(0,4) \dots A(9,86)$; Hydra $C(3,0) \dots C(78,6)$ + conjugacy over 2,000 steps; Fenrir $S(0,1) \dots S(4,35)$	pass
T1	value increase exhaustive to 100,000 + small-case escapes	pass
T2	30,000 randomized checks each for $q = 3$ and $q = 5$; start-independence direct	pass
T3	20,000 identity checks; block runner vs single steps	pass

check	scope	result
T4	minimal-solution iteration to 10^{-6} ; algebraic root exact	pass
deep runs	Hydra 2M / Antihydra 2M / Fenrir 1.5M steps, no halt; drifts match the wiki	pass

Artifacts: `hydra.py` (implementations + wiki fidelity), `theorems.py` (T1–T4 + measurements), `make_hydra_report.py` (this report). Sources: wiki.bbchallenge.org pages Hydra, Antihydra, Hydra function, Fenrir (accessed July 2026); the Coq-BB5 paper for Antihydra’s statement. Status of all three machines: unchanged — open; this report adds proofs beneath the community’s simulations, not a decision.